# Yin Liu

Chapel Hill, NC, 27514 | (404) 451-4185 | Github | yliu6@unc.edu | LinkedIn

## SUMMARY

Survival Analysis; Biopharmaceutical Statistics; Machine Learning & Artificial Intelligence (ML/AI); AI Agents for Statistics; Clinical Trials; Real-World Data & Evidence (RWD/RWE)

## EDUCATION

**University of North Carolina at Chapel Hill** — **Chapel Hill, NC**
*Ph.D. in Biostatistics (Advised by* Drs. Jianwen Cai and Didong Li*)* — 08/2023-05/2028 (Expected)

**University of North Carolina at Chapel Hill** — **Chapel Hill, NC**
*Master in Biostatistics* — 08/2021-05/2023

**Xidian University** — **Shaanxi, China**
*B.S. in Biotechnology (G*raduated 1st class ranking: 1/30*)* — 08/2015-05/2019

## HONORS & AWARDS

| | | |
|---|---|---|
| 2026 | Best Student Paper Award, New England Rare Disease Statistics | Boston, US |
| 2026 | 1st Place of STAI-X Statistical Prediction (Forecasting Accuracy) Award | Boston, US |
| 2026 | 2nd Place of STAI-X Statistical Skill / Agent Module Award | Boston, US |
| 2021 | President's List (Top 5%), Abace Biotechnology Co., Ltd. | China |
| 2019 | Excellent Papers Award (Top 2%), Xidian University | China |
| 2019 | Outstanding Graduation Design (Top 4%), Xidian University | China |
| 2017 | President's List (Top 5%), Shaanxi Shenlong Biological Technology Co., Ltd. | China |
| 2017 | National Scholarship (Top 0.2%, Highest scholarship for undergraduates), Ministry of Education of China | China |
| 2017 | First-class Scholarship (Top 5%), Xidian University | China |

## RESEARCH

**Understanding Overparametrization in Machine Learning Models**

- Built large-scale PyTorch pipelines for DeepSurv, PC-Hazard, MTLR, and NNet models on high-performance computing.
- Conducted hundreds of training configurations to study optimization and generalization in survival setting.
- Characterized how model capacity and data dimensionality shape generalization in survival ML.
- Developed practical guidance for sizing and training survival ML models across low- and high-dimensional settings, linking parameterization to stable out-of-sample performance.

**Generative AI for Reliable Survival Data Synthesis**

- Developed a principled framework of generative models for faithful learning of joint distributions in right-censored survival data.
- Established survival-specific fidelity metrics capturing hazard dynamics and downstream predictive utility.
- Built large-scale PyTorch training pipelines and benchmarked generative models on survival datasets.

**Robust Inference and Survival Prediction for Real-World Data (RWD)**

- Developed a robust inference framework for complex real-world data with non-random sampling.
- Established theoretical properties ensuring calibrated standard errors and confidence intervals.

- Extended the framework to generalized linear models, Cox proportional hazards models for right- and interval-censored outcomes, and frailty models.
- Built a resample–fit–aggregate pipeline to produce Simple-Random-Sample-like subsamples and design-consistent point and variance estimates.
- Developed and evaluated tree-based ensemble ML methods for survival prediction in case-cohort studies.

**Statistical and Artificial Intelligence Methodology for Basket Clinical Trials in Oncology**

- Evaluated Probability of Success metrics across alternative information borrowing methods to characterize basket-level and program-level trial performance.
- Investigated large language model (LLM)-based synthesis of biological evidence to identify biologically similar cancer subtypes and inform downstream statistical borrowing.

**AI-Based Assessment of Post-Surgical Trachoma Outcome**

- Built an end-to-end segmentation-to-classification U-Net based pipeline for post-surgical trachoma image analysis using clinician field-grade labels from >3,000 participants.
- Evaluated segmentation and classification robustness across multiple clinical label sources.

## EXPERIENCE

**UNC Biostatistics Department** **Chapel Hill, NC**
**Teaching Assistant** 08/2025-05/2026 and 08/2023-12/2023

- Teaching Assistant for Causal Inference in Biomedical Research and Principles of Statistical Inference.
- Explained statistical concepts to students from diverse backgrounds and led paper discussions, office hours, and collaborative problem-solving sessions.

**UNC Collaborative Studies Coordinating Center (CSCC) and Department of Epidemiology** **Chapel Hill, NC**
**Graduate Research Assistant** 08/2022-Present

- Developed a statistical analysis plan (SAP) for a Women's Health Initiative observational study, specifying endpoint derivations, regression models, missing-data and sensitivity analyses, expected precision of prevalence estimates, and table and figure shells.
- Analyzed longitudinal data using GEE and mixed-effects models for correlated and clustered outcomes.
- Applied multiple imputation to missing data and evaluated model performance in the Hispanic Community Health Study / Study of Latinos (HCHS/SOL) study.

**Shaanxi Shenlong Biological Technology Co., Ltd.** **Xi'an, China**
**Clinical Trial Research Assistant** 05/2017-08/2017

- Participated in protocol development and statistical analysis for a multicenter Phase III clinical trial.
- Analyzed safety and longitudinal clinical outcomes and prepared statistical results for interim reporting.

## SKILLS

| | |
|---|---|
| **Programming** | R, SAS, Python, C, MATLAB, SQL, Stata, Mplus |
| **Tools** | Bash, Git, SLURM/HPC, Docker, LaTeX |

## COURSEWORK

**Statistical Theory and Methods:** Survival Analysis, Machine Learning, Causal Inference, Computational Biology, Clinical Trial, Longitudinal Data Analysis, Categorical Data Analysis, Survey Sampling, Bayesian Statistics, Stochastic Processes

**Health and Biomedical Sciences:** Epidemiology, Global Public Health, Genetics, Molecular Biology

## PUBLICATIONS & MANUSCRIPS

- **Liu Y.**, Cai J., and Zeng D.(2025+). Inverse Sampling Variance Estimation for M-Estimators in Multi-Stage Complex Designs. *In preparation; target Biometrics*.
- Li, H., Nelson, K., Liu, Y., Yue, H., Shu, Y., Deng, Q., Manepalli, S., **Liu, Y.**, Sullivan, H., Bayly, H., Fan, Y., He, J., Jin, Y., Ruan, Y., Tang, Y., Bellamy, S., and Patil, P. (2026+). Case-Cohort Studies in Practice: A Review to Inform Statistical Research. *In preparation; Target PLOS Global Public Health.*
- Li, H., Fan, Y., He, J., Jin, Y., Ruan, Y., Tang, Y., Liu, Y., Yue, H., Shu, Y., Deng, Q., Manepalli, S., **Liu, Y.**, Sullivan, H., and Bayly, H. (2026+) Review of Statistical Methodology for Case-Cohort Studies. *In preparation; Target Statistical Science.*
- Staggs, A., Wang, T., Tapan, U., **Liu, Y.**, Nelson, K., Agrawal, M., Pu, Y., and Li, H. (2026+). Utilizing Large Language Models to Improve Clinical Decision-Making in Basket Trials. *In preparation; Target BMC Medical Research Methodology.*
- Sullivan, H., **Liu, Y.**, Abrahamson, L., Shu, Y., Li, X., Xu, Z., Lu, W., Liu, M., Li, H. (2026+). Probability of Success in Basket Trials: Evaluation and Comparison of Information Borrowing Methods. *In preparation; Target Journal of Biopharmaceutical Statistics*.
- Zhang, C., Yue, H., **Liu, Y.**, Li, H., and Prince Nelson, S. (2026+). Conditional Inference Survival Forest in Case-Cohort Studies. *Under review; BMC Medical Research Methodology.*
- **Liu Y.**, Li D., and Cai J. (2026+). Understanding Over-parametrization in Survival Models through Interpolation. *Under review; Journal of Machine Learning Research, arXiv: https://arxiv.org/html/2512.12463v3*
- Xu X., Li H., Li K., Zeng Q., and **Liu Y.** (2019), "A Photo-triggered Conjugation Approach for Attaching RGD Ligands to Biodegradable Mesoporous Silica Nanoparticles for the Tumor Fluorescent Imaging", *Nanomedicine: Nanotechnology, Biology and Medicine, Doi: https://doi.org/10.1016/j.nano.2019.04.005*

## PRESENTATIONS

- "Generative AI for Reliable Survival Data Synthesis" *(Invited Seminar; Scheduled) Department of Biostatistics, Boston University*, Boston, MA, 2026.
- "Generative AI for Reliable Survival Data Synthesis" *(Poster; Accepted) Annual New England Rare Disease Statistics (NERDS) Workshop*; Boston, MA, 2026.
- "Generative AI for Reliable Survival Data Synthesis"*(Poster; Accepted) Boston Pharmaceutical Statistics Symposium; Boston, MA, 2026.*
- "Overdose Emergency Department Rate Forecasting with Ensembled Machine Learning" *(Contributed Talk) STAI-X, Harvard University*, Boston, MA, 2026
- "Nanoparticle-Based Fluorescence Imaging for Precision Tumor Detection" *(Student Representative and Award Presenter) China National Scholarship Ceremony*, China, 2017.

## CONFERENCE SHORT COURSE

- "An Overview of Machine Learning and Artificial Intelligence (ML/AI) Methods for Survival Data" *(Scheduled; Joint Short Course with Haolin Li) Duke-Industry Statistics Symposium*, Durham, NC, 2027.